



LaSalle County Station
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10 CFR 50.73

RA18-037

May 21, 2018

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

LaSalle County Station, Unit 1
Renewed Facility Operating Licenses No. NPF-11
NRC Docket No. 50-373

Subject: Licensee Event Report 2018-004-00, Technical Specification Required
Shutdown due to Reactor Pressure Boundary Leakage

In accordance with 10 CFR 50.73(a)(2)(i)(A) and 10 CFR 50.73(a)(2)(ii)(A), Exelon Generation Company, LLC (EGC) is submitting Licensee Event Report (LER) Number 2018-004-00 for LaSalle County Station, Unit 1.

There are no regulatory commitments in this letter. Should you have any questions concerning this report, please contact Mr. Guy V. Ford, Jr., Regulatory Assurance Manager, at (815) 415-2800.

Respectfully,

A handwritten signature in black ink, appearing to read "Harold T. Vinyard", written over a horizontal line.

Harold T. Vinyard
Plant Manager
LaSalle County Station

Enclosure: Licensee Event Report

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – LaSalle County Station



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NE08-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name

LaSalle County Station, Unit 1

2. Docket Number

05000373

3. Page

1 OF 3

4. Title

Technical Specification Required Shutdown due to Reactor Pressure Boundary Leakage

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
03	22	18	2018	- 004	- 00	05	21	18	NA	NA
9. Operating Mode			11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)							
1			☐ 20.2201(b)	☐ 20.2203(a)(3)(i)		☒ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		
			☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		
			☐ 20.2203(a)(1)	☐ 20.2203(a)(4)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		
			☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		
10. Power Level			☐ 20.2203(a)(2)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(v)(A)		☐ 73.71(a)(4)	
24			☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)		☐ 50.73(a)(2)(v)(B)		☐ 73.71(a)(5)		
			☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(1)		
			☐ 20.2203(a)(2)(v)	☒ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(i)		
			☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(vii)		☐ 73.77(a)(2)(ii)		
			☐ 50.73(a)(2)(i)(C)		☐ Other (Specify in Abstract below or in NRC Form 366A)					

12. Licensee Contact for this LER

Licensee Contact

Jeff Stovall, Operations Director

Telephone Number (Include Area Code)

(815) 415-3100

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
X	AD	V	A585	Yes	NA	NA	NA	NA	NA

14. Supplemental Report Expected

☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No

15. Expected Submission Date

Month	Day	Year
NA	NA	NA

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On March 22, 2018, a through-wall pipe leak was identified on a three-fourths inch vent line coming off the bonnet on a reactor recirculation pump discharge valve. This condition was reactor coolant system (RCS) pressure boundary leakage, which required entry into technical specifications (TS) actions for required shutdown. The leakage was significantly less than the 10 gallons per minute emergency action threshold. At the time of discovery, Unit 1 was in Mode 1. A controlled shutdown of Unit 1 was performed, having all necessary shutdown equipment available; and, there was no impact to Unit 2. This condition was reported on March 22, 2018 (ENS 53276) in accordance with 10 CFR 50.72(b)(2)(i) for plant shutdown required by Technical Specifications and 10 CFR 50.72(b)(3)(ii)A for the pressure boundary leakage as a principal safety barrier being in a degraded condition.

The cause for the steam leak on the vent line was determined to be fatigue cracking that initiated on the external surface of the pipe at the toe of the socket fillet weld. Crack propagation occurred under low cycle fatigue conditions. In addition, the fracture surface contained deformation features which were indicative of a potential high nominal stress. Actions were taken to repair the vent line after plant shutdown and perform failure analysis on the degraded component. This condition is reportable in accordance with 10 CFR 50.73(a)(2)(i)A as a condition that required a plant shutdown in accordance with TS and 10 CFR 50.73(a)(2)(ii)A as a principal safety barrier being in a degraded condition.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
LaSalle County Station, Unit 1	05000373	2018	- 004	- 00

NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

LaSalle County Station Unit 1 is a General Electric Boiling Water Reactor with 3546 Megawatts Thermal Rated Core Power.

The affected component was a three-fourths inch vent line (1RR32AB-3/4") coming off the bonnet on the 1B reactor recirculation pump discharge valve (1B33-F067B).

CONDITION PRIOR TO EVENT

Unit(s): 1	Date: March 22, 2018	Time: 0300 CDT
Reactor Mode(s): 1	Mode(s) Name: Power Operation	Power Level: 24 percent

DESCRIPTION

On March 22, 2018, at 0300 CDT, a through-wall pipe leak was identified on a three-fourths inch vent line coming off the bonnet on the 1B reactor recirculation pump discharge valve (1B33-F067B). Personnel observed a plume of steam was coming from a partial circumferential crack at the toe of the weld connecting the small-bore vent pipe 1RR32AB-3/4" and the valve bonnet.

This condition was determined as a reactor coolant system (RCS) pressure boundary leakage, which required entry into technical specifications (TS) actions for required shutdown. Specifically, TS 3.4.5, Reactor Coolant System Operational Leakage, Required Actions C.1 and C.2 were entered, which require the unit to be in Mode 3, Hot Shutdown within 12 hours, and Mode 4, Cold Shutdown in 36 hours, respectively.

The leakage was significantly less than the 10 gallons per minute emergency action threshold. At the time of discovery, Unit 1 was in Mode 1. A controlled shutdown of Unit 1 was performed, having all necessary shutdown equipment available. Unit 1 was placed into Mode 4 on March 22, 2018, at 1652 CDT. There was no impact to Unit 2. This condition was reported to the NRC via Event Notification System (ENS) 53276 on March 22, 2018 as a degraded condition that involved a required plant shutdown.

CAUSE

The failure occurred after approximately seven days of service following the vent line installation during the Unit 1 refueling outage (L1R17). A component failure analysis was completed which determined that the failure was attributed to fatigue cracking that initiated on the external surface of the pipe at the toe of the socket fillet weld. Based on the relatively large spacing of the fracture surface fatigues striations, crack propagation occurred under low cycle fatigue conditions. In addition, the fracture surface contained deformation features which were indicative of a potential high nominal stress.

For the failed "B" vent line assembly, the likely cause for the high nominal stress was contact with the clamshell support clamp. Based on the clamshell clamp and pipe wear marks, the support appeared to have applied a downward force to the failed line during service. Lab inspections did not detect fatigue cracking at the socket weld location in the similar intact "A" vent line, which suggests that failure was not imminent.

The station initiated a root cause investigation to determine additional causal factors for this failure.

REPORTABILITY AND SAFETY ANALYSIS

This condition is reportable in accordance with 10 CFR 50.73(a)(2)(i)A as a condition that required a plant shutdown in accordance with TS and 10 CFR 50.73(a)(2)(ii)A as a principal safety barrier being in a degraded condition. This condition was reported to the NRC via ENS report number 53276 on March 22, 2018.

There were no safety consequences resulting from the event. Makeup capability was adequate to compensate for the leak. All emergency core cooling systems (ECCS) were operable and capable of fulfilling their intended safety functions during the period of leakage. The leak was within the capability of the leak detection system and was within the capability of the leak collection system.

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CONTINUATION SHEET**

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NARRATIVE

The event did not constitute a safety system function failure (SSFF) as defined in accordance with NEI 99-02, "Regulatory Assessment Performance Indicator Guideline."

CORRECTIVE ACTIONS

Immediate corrective actions taken in response to the condition were:

- Performed required plant shutdown
- Removed the degraded vent line and valves from 1B33-F067B
- Removed the vent lines and valves from 1B33-F067A
- Performed failure analysis of the failed component
- Initiated a causal investigation

Additional corrective actions may be determined from the root cause investigation.

PREVIOUS OCCURRENCES

LER 374-2015-003-00

On August 7, 2015, LaSalle Unit 2 was in Mode 3 for a planned maintenance outage. During the initial drywell entry, a steam leak was observed on the reactor recirculation (RR) system line 2RR94AB-3/4", which is upstream of valve 2B33-F080B (RR Pump Discharge Valve 2B33-F067B Inspection Port - Reactor Side Upstream Stop Valve). The leak was determined to be pressure boundary leakage. Technical Specification 3.4.5, "RCS Operational Leakage," Required Actions C.1 and C.2 were entered, which required the unit to be in Mode 3 within 12 hours and in Mode 4 in 36 hours, respectively. The cause of the steam leak was determined to be poor weld quality and vibration induced fatigue. The weld was repaired during the maintenance outage.

COMPONENT FAILURE DATA

Manufacturer: Atwood & Morrill (A585)
Device: Reactor Recirculation Pump Discharge Valve 1B33-F067B
Component ID: Model Series 9085 Wedge Gate Valve